Lab 9- Filled In

April 9, 2026 10:10 AM

Practice Problems: Two populations Mean and Proportion Difference Test

1. A team tests whether a new debugging tool reduces the number of errors in code. Use the provided R output to perform all steps of the test.

before = c(18, 20, 16, 22, 19, 21, 17, 20)	shapiro.test(before)	shapiro.test(after)
after = c(15, 18, 14, 19, 16, 18, 15, 17)	data: before	data: after
d = before – after >0	W = 0.97365, p-value = 0.925	W = 0.94352, p-value = 0.646
$\boxed{d = 3\ 2\ 2\ 3\ 3\ 3\ 2\ 3}$	H_0: The underlying distribution	is normal
sd(d) = 0.5175492	H_A: The underlying distribution	is not normal
paired t-test		

qt(0.05,7) = -1.894579 qt(0.025,7) = -2.364624

$$t = \frac{\bar{d} - \mu_d}{\frac{S_d}{\sqrt{n}}}$$

H_0: $\mu_d = 0$

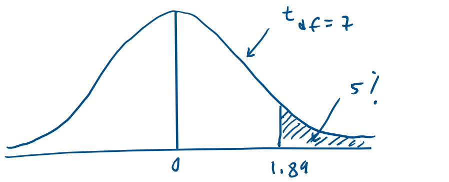
H_A: $\mu_d > 0$

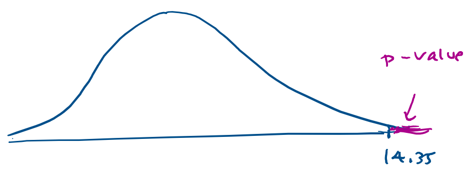
This is a right-tailed paired t-test

$$\bar{d} = \frac{21}{8} = 2.625$$

$$sd = \sqrt{\left(\left((3-2.625)^2 + (2-2.625)^2 + \dots + (3-2.625)^2\right)/7\right)}$$

$$t = \frac{2.625 - 0}{\left(\frac{0.5175492}{\sqrt{8}}\right)} = 14.35$$





Decision: P-value is less than 5% (default α)

we reject Ho

Conclusion: At 5% significance level the evidence suggests that the true mean bug count is reduced.

Note: Both "before" and "after" sample data seem to come from normal distributions.

2. A company compares response times (ms) between two cloud providers to detect mean difference. Use the provided output to perform the test manually.

Independent, two sample, population mean t-test

```
ProviderA = c(110, 115, 113, 130, 120, 118, 117)

ProviderB = c( 99, 100, 105, 98, 102, 99, 111)

t.test(ProviderA,ProviderB, alternative = "less")


        Welch Two Sample t-test

data:  ProviderA and ProviderB

t = ?, df = 10.917, p-value = 0.9999

alternative hypothesis: true difference in means is less than 0

sample estimates:

mean of x mean of y

 117.5714  102.0000
```

$$t = \frac{(\bar{x}_1-\bar{x}_2)-(\mu_1-\mu_2)}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$$

$$t = \frac{(117.5714-102)-0}{\sqrt{\left(\frac{6.4^2}{7}+\frac{4.62^2}{7}\right)}} = \frac{15.57}{2.983} = \boxed{5.22}$$

```
sd(ProviderA) = 6.399405
sd(ProviderB) = 4.618802
```

6.4

```
shapiro.test(ProviderA)

data:  ProviderA

W = 0.91651, p-value = 0.4428
```

Normal ✓

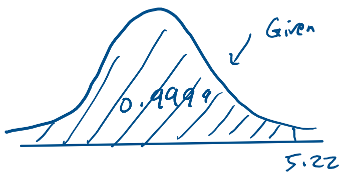
```
shapiro.test(ProviderB)

data:  ProviderB

W = 0.83558, p-value = 0.09034
```

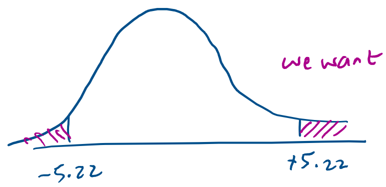
Normal ✓

$H_0: \mu_A - \mu_B = 0$

$H_A: \mu_A - \mu_B \neq 0$

} two-tailed test





p-value $= 2(1-0.9999) = \boxed{0.0002}$

Decision: P-value is much less than alpha, we reject H0

Conclusion: at a 5% significance level, the evidence suggests that there is a difference between the true mean response times of the two provided cloud services.

3. A company tests two app versions:

- Version A: 180 out of 300 users enabled a feature
- Version B: 210 out of 320 users enabled a feature

Is the success rate in one version higher than the other?

$H_0: P_A - P_B = 0$

$H_A: P_A - P_B \neq 0$

This is a two-tailed proportion difference test of two independent populations.

$\hat{p}_1 = \frac{180}{300} = 0.6 \qquad \hat{q}_1 = 0.4$

$\hat{p}_2 = \frac{210}{320} = 0.65625 \qquad \hat{q}_2 = 0.34375$

? $n_1\hat{p}_1 \geq 10 \qquad n_1\hat{q}_1 \geq 10$ ✓

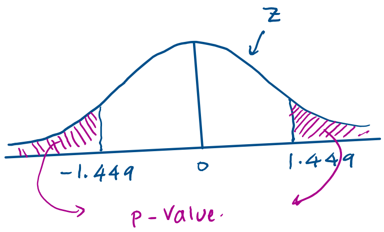
? $n_2\hat{p}_2 \geq 10 \qquad n_2\hat{q}_2 \geq 10$ ✓

Normal? Yes

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2} = \frac{180 + 210}{300 + 320} = 0.629$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (P_1 - P_2)}{\sqrt{\left(\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right)}} = \frac{(0.6 - 0.65625) - 0}{\sqrt{\left(0.629 \times 0.371 \times \left(\frac{1}{300} + \frac{1}{320}\right)\right)}} = -1.449$$

$$\text{p-value} = 2\left(\text{pnorm}(-1.449)\right)$$



According to the 68-95-99% rule for standard normal distribution, the p-value is greater than 5%.

Decision: we fail to reject H_0

Conclusion: the population proportion of enabling the feature is not significantly different is these two versions of the app.

4. A systems engineer wants to study whether server load (%) affects response time (ms). Perform the test manually.

Server Load (%)	Response Time (ms)
10	120
20	130
30	145
40	160
50	175
60	190
70	210
80	230
90	250
95	270

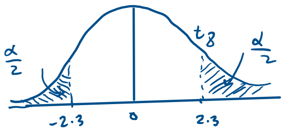
Given: $r = 0.9926386$ $n = 10$

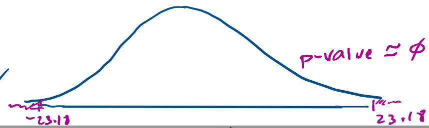
$H_0: \rho = 0$ two-tailed test

$H_A: \rho \neq 0$

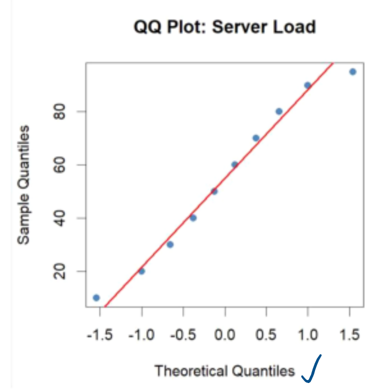
$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9926386\sqrt{8}}{\sqrt{(1-0.9926386^2)}} = 23.18$$

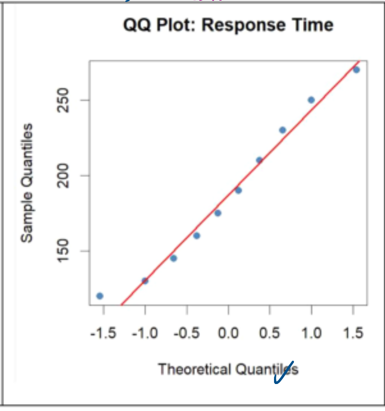
Given:



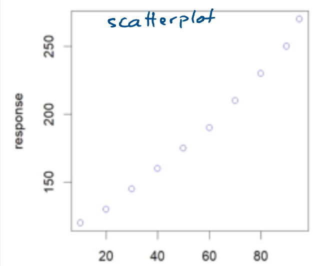


Normality Check ✓





cor = 0.9926386
qt(0.025,8) = -2.306004
qt(0.05,8) = -1.859548



No outliers ✓
linear trend ✓

Decision: the p-value is almost zero, we have strong support for rejecting Ho

Conclusion: there is a strong enough positive linear relation between server load and response time